

Surya Attendance & Payroll Management System

End-to-End Implementation Report & Architecture Specification

1. Executive Summary

The **Surya Attendance & Payroll Management System** is a lightweight, secure Single Page Application (SPA) designed for employee time-tracking, shift schedules, geo-verification, and payroll processing. Built for **HS Group Delhi (House of Surya)**, the application facilitates employee check-ins, shift swap request coordination, profile database management, and payroll statements, with strong role-based access control (RBAC).

2. System Architecture

The application runs as a front-end SPA supported by a mocked database layer stored in browser `localStorage`.

- **Routing Layer (`js/app.js`)**: A client-side hash router that parses navigation links (e.g., `#dashboard`, `#admin-schedules`, `#employee-swaps`) and mounts views dynamically.
- **Database & Calculation Layer (`js/db.js`)**: An object-oriented API that performs schema queries, database transactions, shift assignments, and payroll calculations.
- **Security & Authentication (`js/auth.js`)**: Validates passwords, tracks active sessions, and simulates biometric registers.

3. Phased Implementation Details

Phase 1: Location & Geofencing System Overhaul

- **Mock Office Geofence Coordinates:**
 1. **Kohat Enclave, Pitampura, Delhi (HQ)** — Coord: `28.6978° N, 77.1408° E` (In-Range).
 2. **Chandni Chowk (Hub)** — Coord: `28.6562° N, 77.2310° E` (Out of Range - 9.8 km deviation).
 3. **Omaxe City, Delhi (Branch)** — Coord: `28.8130° N, 77.0673° E` (Out of Range - 14.5 km deviation).
- **Clock-in Verification Logic:**
 - Biometric actions (Face ID, Fingerprint) are restricted to the HQ location.
 - Attempting to clock in from an out-of-range location disables biometric buttons.
 - Employees can check in via Passcode only after filling out an **Out-of-Geofence Deviation Justification** textarea.

- Out-of-range logs are flagged as `Pending Manager Review`.

Phase 2: Shift Schedule Location Selector

- **Shift Settings Modal:** A location dropdown selection was added to the **Create / Edit Shift Calendar** modal.
- **Inline Shift Card Controls:** Added interactive **Location Select** and **Location Swap** dropdown selectors directly onto the shift cards on the admin page.
- Changing location instantly saves modifications to `localStorage`.
- Selecting another shift from the swap dropdown swaps the locations between both patterns immediately.

Phase 3: Location-Aware Shift Swapping

- **Coworker Preview:** The coworker swap dropdown displays each coworker's assigned shift name and location.
- **Coworker Assignment Preview Block:** Selecting a coworker displays a detailed current assignment card showing their shift times and location before submission.
- **Swap Mode Configurations:**
 1. **Swap Both Shift & Work Location:** Exchanges the shift timings and the preferred work locations of both users.
 2. **Swap Work Location Only:** Exchanges the preferred work locations while keeping their current shift timings.
 3. **Swap Shift Only:** Exchanges shift timings while keeping their current preferred work location.
- **Ledgers Update:** Updated the Sent Requests, Received Requests, and Manager Approval logs to display locations and swap modes.

Phase 4: Role-Based Payroll & Profile Access

- **Segregation of Duties:** Both the **HR Coordinator** and **Operations Manager** can now access, edit, and configure the payroll/salaries of general Employees (role === 'employee').
- **HR Payroll Controlled by Manager:**
 - Operations Manager can view, edit, and delete the HR Coordinator's profile and payroll.
 - The HR Coordinator is hidden from the register table when logged in as HR, preventing them from modifying their own salary, base payroll coefficients, or portal access roles.
- **Role Escalation Protection:** The user editor modal restricts portal role assignments; only the Operations Manager can change user access roles (assigning `Employee`, `HR Coordinator`, or `Operations Manager`). When edited by HR, the selection is restricted to the `Employee` role.

Phase 5: Manager Payslip Adjustments

- **Interactive Adjustments Drawer:** Added an **Edit Adjustments** button to the Admin Payslip Preview drawer (visible only to the logged-in Operations Manager).
- **Adjustment Modals:** Clicking this button opens a modal to customize:
 - **Discretionary Bonus** (Earning Addition)

- **Ad-hoc Deductions** (Deduction Subtraction)
- **Manager Remarks / Notes** (Printed directly onto the payslip statement)
- **Calculations & Persistence:** Integrated adjustments directly into the database calculations (`js/db.js`). The net disbursed payouts reflect the overridden values immediately.
- **Payslip Template Update:** Modified both the administrator inspect template and the employee's personal payslip statement (`js/app.js`) to display discretionary bonuses, custom deductions, and remarks dynamically when present.

Phase 6: Real-Time Geofencing & Native WebAuthn (Passkey) Integration

- **Dynamic Geofence Verification:**
 - Added "**Use Device GPS (Real-Time Location)**" toggle to allow employees to request their actual location via `navigator.geolocation.getCurrentPosition``.
 - Computes exact real-time distance using the **Haversine Formula** (geodesic distance on a spherical earth) against their *assigned shift location*.*
 - Restricts attendance logging: if an employee is outside the 100-meter radius, biometric clocking is disabled, and manual clock-in requires an **Out-of-Geofence Deviation Justification** (pending manager review).
- **Native WebAuthn Biometric Authentication:**
 - Integrated the W3C **WebAuthn API** (Passkeys) for secure, platform-bound biometric credentials (Touch ID, Face ID, Windows Hello).
 - Employees can enroll their hardware biometrics by calling `navigator.credentials.create()``, which registers a secure public-key credential on their device.
 - Clocking in and out requests authentication via `navigator.credentials.get()``.
 - Fully supports **graceful degradation**: if WebAuthn is unsupported (e.g., non-secure context) or permission is denied/cancelled, the application seamlessly falls back to the customized visual scanning simulators (webcam Face HUD or Fingerprint scanpad).

4. Technology Deep Dive: Geofencing & Biometrics

A. Geofencing & Haversine Distance Calculation

Geofencing restricts employee check-ins to predefined coordinate boundaries. The distance between the employee's coordinates and the assigned workplace is calculated using the **Haversine Formula**:

- $d = 2 * R * \arcsin(\sqrt{\sin^2(D_lat / 2) + \cos(lat1) * \cos(lat2) * \sin^2(D_lon / 2)})$
- Where:
 - **d** is the distance between coordinates in meters.
 - **R** is the Earth's radius (approx. 6,371,000 meters).
 - **lat1, lat2** are the latitudes of both points in radians.
 - **D_lat, D_lon** are the differences in latitude and longitude in radians.

- If $d \leq 100$ meters, the employee is "In Range" and biometric clock-in is enabled. If $d > 100$ meters, they are flagged for geofence deviation.

B. W3C WebAuthn Biometric Authentication

WebAuthn (Web Authentication) is a web standard that allows servers to authenticate users using public-key cryptography instead of passwords.

- **Initialization / Registration:**

- The browser requests credential creation via `navigator.credentials.create()`.
- The OS authenticates the user (via fingerprint, face scan, or PIN) and generates a unique cryptographic key pair.
- The public key and credential ID are returned and stored in the database (`localStorage`), while the private key remains secure in the device's hardware TPM/Enclave.

- **Verification / Authentication:**

- When checking in, the browser requests assertion via `navigator.credentials.get()` matching the stored credential ID.
 - The user scans their biometric, signing a server challenge using their private key.
 - The app validates the signature response, granting instant access without requiring manual passwords.
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5. System Setup & Initialization Guide

Running the App Locally

The application is served locally via a Python HTTP server on port 8012:

```
```bash
python -m http.server 8012
```
```

To run the server, browse to: **<http://localhost:8012>**

Biometric Setup Initialization

1. Log in as any employee (e.g., username `john`, password `JohnPassword123!`).
2. Click **Setup Face ID** or **Setup Fingerprint** on the top warning banner.
3. Choose **Yes** to register your device's native hardware credentials. Authenticate via your OS prompt.
4. If your device doesn't support platform authenticators or you choose to skip, the system will initialize the simulated enrollment scanner, allowing you to complete setup.

Geofencing Simulation & Real Mode

1. In the **Simulated GPS Geofence** card, select **Use Device GPS (Real-Time Location)**.
2. Grant the browser permission to access your location.

3. The app will fetch your real-world coordinates, compute the distance to your assigned workplace, and enforce geofence policy constraints immediately.

Simulated User Credentials

1. **Operations Manager** (Full Control, edit HR, adjust payslips):

- Username: `manager`
- Password: `ManagerPassword123!`

2. **HR Coordinator** (Manage employee shifts & profiles, no self-edits):

- Username: `hr`
- Password: `HRPassword123!`

3. **General Employee** (John Doe):

- Username: `john`
- Password: `JohnPassword123!`